data availability
open data
data on request
closed data
preprint
reanalysis on open data

decoding mode
online
offline

electrode type
ECoG
Utah array
thin-film threads
microwire array

speech ability
participants who can speak naturally
participants who have vocal paralysis

task instruction
inner speech
overt / mouthing / intended

decoded output
brain-to-voice
brain-to-text

## Speech

**Boccato et al., 2026 ◆/◇**
cross-subject CTC ⬡, WER 10.3%/5.9%, vocab 125k (n=4, T12△/T15△/T16△/T17△)

**Y. Zhang et al., 2026 ◆/◇**
BrainTo-Text (BIT) ⬡, WER 5.10%/1.76% (E2E 10.22%/7.76%), vocab 125k (n=2, T12△/T15△)

**Pouratian et al., 2026 ◇**
fully implanted speech BCI ⬡, 75 wpm, WER 5.8%, vocab 125k (n=3, V1△/V2△/V3△)

**Card et al., 2024 ◇**
rapid calibration ⬡, WER 4.9%, vocab 125k (n=1, T15△)

**Feghhi et al., 2025 ◇**
time-masked Transformer ⬡, WER 7.98% (ens. 5.68%), vocab 125k (n=1, T12△)

**Paradromics, 2026 ◇**
real-time speech ⬡, WER n/r, vocab n/r (n=1, C1△)

**Willett et al., 2023 ◇**
large-vocab decoding ⬡, 62 wpm, WER 23.9%, vocab 125k (n=1, T12△)

**Littlejohn et al., 2025 ◇**
streaming voice ⬟, listener WER 58.8%, vocab 1,024 (n=1, B3△)

**Yoon et al., 2026 ◇**
real-time ensembles ⬡, WER 33.7→26.0%, vocab 125k (n=1, T12△)

**Z. Zhang et al., 2026 ◇**
ALIGN cross-day ⬡, unseen-day WER 32.2%/45.5%, vocab 125k (n=2, T12△/T15△)

**Metzger et al., 2022 ◇**
NATO code-word spelling ⬡; WER 10.5%, vocab 1,152 (n=1, B1△)

**Wandelt et al., 2024 ◆/◇**
internal speech ⬡, WER 21%/77%, vocab 8 (n=2, P1▲/P2▲)

**Kunz et al., 2025 ◆/◇**
inner speech ⬡, WER 23.7%/51.3%/46.6%, vocab 125k (n=3, T12△/T15△/T16△)

**Wairagkar et al., 2026 ◇**
Brain2voice 2.0 ⬟, listener WER 5.24%, open vocab (n=1, T15△)

**n = 16 unique participants**

**Moses et al., 2021 ◇**
real-time sentence decoding ⬡, 15.2 wpm, WER 25.6%, vocab 50 (n=1, B1△)

**Metzger et al., 2023 ◇**
78 wpm + avatar ⬟/⬡, WER 25.5%, vocab 1,024 (n=1, B3△)

**Wairagkar et al., 2025 ◇**
voice synthesis ⬟, listener WER 43.8%, open vocab (n=1, T15△)

**Jude et al., 2026b ◆/◇**
intended speech in LIS ⬡, WER 66.5%, vocab 125k (n=1, T17△)

**Card et al., 2026 ◇**
3,800 h home ⬡, WER 0.8-3.5%, vocab 125k (n=1, T15△)

**Angrick et al., 2024 ◇**
speech synthesis ⬟, listener WER 20%, vocab 6 (n=1, CC01△)

**Li et al., 2025 ◇**
DConD-LIFT ⬡, WER 5.77% (1-model 8.06%), vocab 125k (n=1, T12△)

**Fogg et al., 2026 ◇**
multi-user ⬡, WER 3.9-8.0%, vocab 125k (n=6, T12△/T15△/T16△/T17△/T21△/T22△)

**Silva et al., 2024b ◇**
bilingual ⬡, WER 25.0%, vocab 178 (n=1, B1△)

**Levin et al., 2026 ◇**
cross-brain transfer ⬡, PER 23-40% (WER n/r), vocab 125k (n=3, T12△/T15△/T16△)

**Brosler et al., 2026 ◇**
speech+gesture avatar ⬡, WER 30.0%, vocab 5-10 (n=3, B1r△/B3△/B6△)

**Feng et al., 2024 ◇**
E2E RNN-LLM ⬡, WER 26.3% (B2T 24.69%), vocab 125k (n=1, T12△)

**Nason-Tomaszewski et al., 2026 ◇**
Q&A 27.7 wpm ⬡, WER 35.2%, vocab 125k (n=1, T16△)

**Feghhi et al., 2026 ◇**
LightBeam ⬡, WER 5.01%/2.36%, vocab 100k (n=2, T12△/T15△)

## Typing

**Neuralink, 2026 ◇**
ten-finger typing ⬡, 40 wpm, WER n/r, vocab n/r (n=1, P7▲)

**Jude et al., 2026a ◇**
touch typing neuroprosthesis ⬡, 110 cpm, WER 12.5%/3.4%, vocab 125k (n=2, T17△/T18▲)

**n = 3 unique participants**

## Handwriting

**Karpowicz et al., 2024 ◇**
FALCON H2 benchmark ⬡, WER 11.0% (held-in 2.0%), vocab 130k (n=1, T5▲)

**Levin et al., 2026 ◇**
cross-brain transfer ⬡; CER 2% (WER n/r), open vocab (n=3, T5▲/T12△/T18▲)

**Willett et al., 2021 ◇**
brain-to-text ⬡, 90 cpm, WER 25.1%, open vocab (n=1, T5▲)

**Fan et al., 2023 ◇**
403-day plug-and-play ⬡, WER 4.0%, vocab 130k (n=1, T5▲)

**Xu et al., 2025 ◇**
video-cued trajectory decoding ⬡, CER 8.9% (WER n/r), vocab 1,000 (n=1, Z1▲)

**n = 4 unique participants**

2021 2022 2023 2024 2025 2026 2027

*20 unique participants in total; 1 used a system at home (T15)*